RAG

9 RAG Architectures Compared

From beginner Standard RAG to advanced Multimodal — when to use which

Standard RAG

FLOW

User Query → Query Processing → Retrieval → Document Selection → Context Integration → LLM Response

IMPLEMENTATION TIP

Start with smaller chunk sizes (e.g. 256–512 tokens) and experiment with overlap to improve coverage without overwhelming the context window.

WHEN TO USE

Best for straightforward fact retrieval where accuracy matters but complexity is low.

REAL-WORLD EXAMPLE

FAQ bots that surface up-to-date company policies or knowledge base entries.

DeepRAG (Hierarchical + Process Supervision)

FLOW

User Query → Question Decomposition / Sub-questions → Retrieval per sub-question → Reasoning + Verification → Final Response

IMPLEMENTATION TIP

Use hierarchical indicators to manage nested dependencies, and combine process supervision (reward signals) to align sub-queries with final answers.

WHEN TO USE

Multi-hop or domain-specific (e.g. biomedical) questions needing structured decomposition.

REAL-WORLD EXAMPLE

Biomedical QA over linked concept databases (e.g. UMLS).

MA-RAG (Multi-Agent RAG)

FLOW

User Query → Planner, Extractor, Retriever, Synthesizer → Combined Retrieval & Reasoning → LLM Output

IMPLEMENTATION TIP

Each agent handles a subtask, and only the necessary agents are invoked per query. Agents use chain-of-thought prompting (reasoning traces) for interpretability.

WHEN TO USE

Ambiguous or complex queries where decomposition and agent collaboration helps.

REAL-WORLD EXAMPLE

Open-domain QA across heterogeneous sources.

Corrective RAG

FLOW

User Query → Initial Response → Error Detection → Retrieval → Response Correction → Final Response

IMPLEMENTATION TIP

Use a lightweight verification loop — either a separate classifier or an auxiliary LLM — to flag uncertain segments and trigger corrective retrieval.

WHEN TO USE

Suitable when outputs must be double-checked before being shown to the user.

REAL-WORLD EXAMPLE

Clinical decision-support assistants that cross-verify against trusted medical literature.

Speculative RAG

FLOW

User Query → Small Model Draft → Parallel Retrievals → Large Model Verification → Final Response

IMPLEMENTATION TIP

Pair a smaller, faster model for drafting with a larger model that validates against retrieved documents; adjust retrieval parallelism based on latency budgets.

WHEN TO USE

Ideal when response speed is crucial but correctness can't be sacrificed.

REAL-WORLD EXAMPLE

Live chat agents for e-commerce that need rapid draft answers while verifying product details.

Fusion RAG

FLOW

User Query → Multiple Query Generation →

Multi-Retrieval → Fusion of Retrieved Documents →

LLM Response

IMPLEMENTATION TIP

Employ a fusion method like Reciprocal Rank Fusion to merge results, plus weighting sources according to reliability and domain relevance.

WHEN TO USE

Useful when queries require pulling evidence from different types of sources or formats.

REAL-WORLD EXAMPLE

Academic research assistants synthesizing across journal articles, patents, and technical blogs.

RAG-Gym / Process-Supervised Agentic RAG

FLOW

User Query → Agentic Planning / Reasoning →

Step-by-step Retrieval / Search Agents →

LLM Generation + Process Supervision

IMPLEMENTATION TIP

Use process-level rewards / supervision to train agents (e.g. reward which retrieval or reformulation step is better), not just supervise the final answer. The RAG-Gym / ReSearch architecture is from a recent arXiv paper.

WHEN TO USE

For multi-step reasoning or tasks that benefit from supervising intermediate steps.

REAL-WORLD EXAMPLE

QA systems that have to plan a search path rather than one-shot retrieval.

Modular RAG

FLOW

User Query → Module Routing / Scheduling → Retrieval / Rewriting / Fusion via Modules → LLM Generation

IMPLEMENTATION TIP

Decompose into operators (e.g. routing, query rewriting, fusion) so you can replace e.g. your reranker or query reformulator independently. The Modular RAG paper identifies patterns like linear, conditional, branching, looping.

WHEN TO USE

When you want flexibility in customizing or swapping parts of the RAG pipeline.

REAL-WORLD EXAMPLE

Research systems exploring new retrieval/fusion strategies.

Self-adaptive Multimodal RAG (SAM-RAG)

FLOW

User Query (text / multi-modal) →

Adaptive Retrieval Across Modalities →

Filtering & Quality Check → LLM Response

IMPLEMENTATION TIP

Dynamically select which modalities to retrieve from, verify the relevance of retrieved multimodal data, and integrate them in context. The model adapts retrieval count and modality selection per query.

WHEN TO USE

When queries or data involve images, audio, or mixed modalities.

REAL-WORLD EXAMPLE

Q&A combining text + images (e.g. product descriptions + photos).